VERITAS - Digital Alibi Verification System

Course: CS-351 Artificial Intelligence

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1. Abstract

Veritas is a forensic intelligence tool designed to automate the validation of digital alibis. In cybersecurity investigations, determining if a suspect was physically present at a workstation during a specific time window is often a manual and error-prone process. This project presents an artificial intelligence (AI) system that parses raw system logs, extracts behavioral features, and applies a Random Forest Classifier to distinguish between active human presence and passive system noise. By incorporating SHAP (SHapley Additive exPlanations), Veritas makes "black box" predictions transparent by providing evidence for the specific artifacts (e.g., Logons, Privilege Escalation) that lead to the verdict [1].

2. Introduction & Motivation

Digital forensics often depends on establishing a "Timeline of Activity." Investigators must answer critical questions: Was User X at their computer at 9:00 PM, or was the machine running background updates? Current methods rely heavily on manual keyword searching (grep/Ctrl+F), which frequently misses complex behavioral patterns. Our motivation was to create an intelligent agent capable of parsing these logs to reconstruct a user's timeline and produce a probabilistic judgment of their presence, addressing the critical need for automated and scalable forensic tools [2].

3. Problem Definition & Objectives

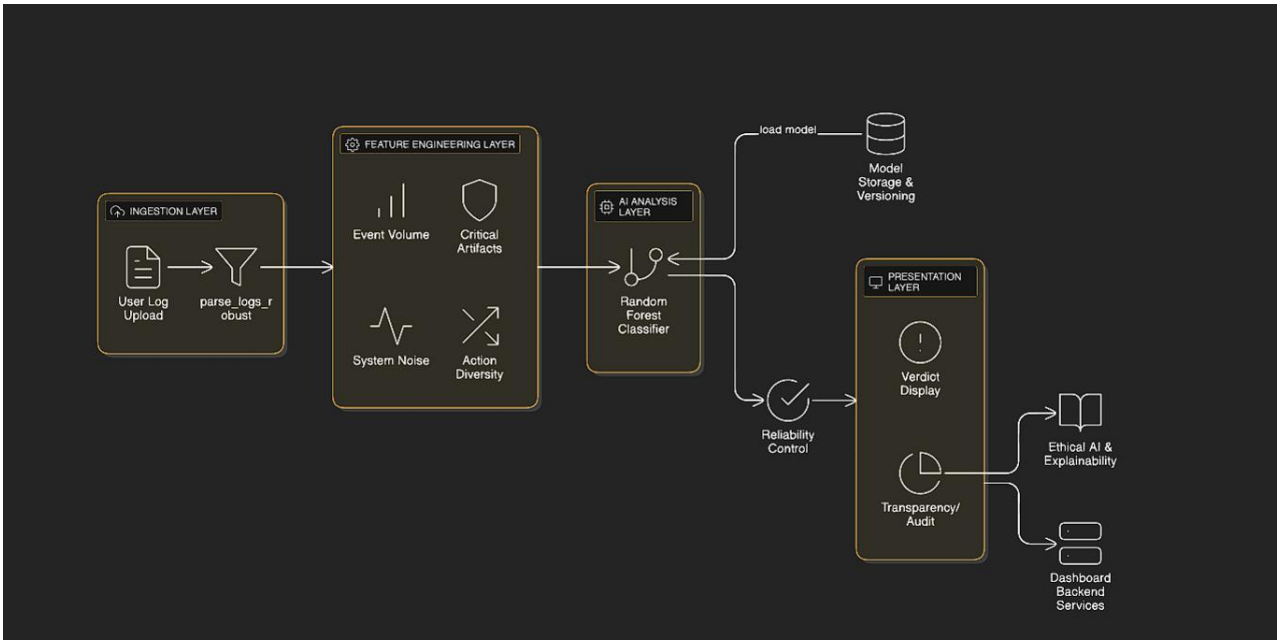
The core problem addressed is "Alibi Verification": Given a claimed time window (e.g., "I was at dinner from 19:00 to 22:00") and a set of system logs, the system must determine if the digital evidence contradicts the claim.

- **Objective 1:** Robustly ingest and filter heterogeneous log formats (PowerShell, CSV, TXT).
- **Objective 2:** Distinguish between human activity (files opened, emails sent) and system noise (updates, telemetry).
- **Objective 3:** Provide explainable, visual proof for the final verdict using SHAP [3].

4. Literature Review

Traditional forensic timeline analysis tools, such as Plaso or Log2Timeline, focus primarily on aggregation—ordering events chronologically—but lack attribution capabilities. Research by the Canadian Institute for Cybersecurity (CIC) suggests that network traffic and log analysis benefit significantly from Ensemble Learning methods like Random Forest due to their ability to handle high-dimensional, noisy data, as demonstrated with the CICIDS2017 dataset [4]. Our approach builds on this foundation by integrating an "Explainability Layer," a feature often missing in commercial forensic AI solutions.

5. System Architecture



The system follows a modular pipeline architecture designed for scalability and interpretability:

1. **Ingestion Layer:** A robust parser (`parse_logs_robust`) reads raw text/CSV logs and filters them strictly by the specific alibi time window.
2. **Feature Engineering:** Raw logs are converted into five forensic feature vectors:
    - **Event Volume:** Total count of non-noise events.
    - **Critical Artifacts:** Specific high-fidelity IDs (e.g., 4624 Logon, 4672 Admin Privileges).
    - **Action Diversity:** Variance in event types to differentiate between automated bots and human actors.
    - **System Noise:** Count of ignored background processes.
3. **AI Analysis Layer:** A pretrained Random Forest Classifier predicts the probability of "Presence."
4. **Heuristic Override Layer:** A rule-based logic gate that forces a "POSITIVE" verdict if definitive proof (like a login event) is found, ensuring critical evidence is never overridden by the ML model.
5. **Presentation Layer:** A Streamlit dashboard displaying the final verdict and SHAP explanation plots.

6. Data Description & Preprocessing

We utilized a hybrid dataset strategy to train and validate the system:

- **Primary Data:** Synthetic logs generated to mimic realistic user behavior (heavy usage vs. stealth usage) combined with realistic system noise.
- **Preprocessing:**
  - **Time Windowing:** Strict filtering to the exact second of the claimed alibi to ensure relevance.
  - **Noise Filtering:** Removal of known background processes (e.g., `igccservice`, `BTHUSB`) using a predefined blocklist.
  - **Normalization:** Features are scaled using `StandardScaler` to ensure the machine learning model treats low-volume "stealth" events fairly compared to high-volume system processes [5].

7. Algorithmic Implementation

- **Machine Learning:** We selected a **Random Forest Classifier** (`n_estimators=100`, `max_depth=8`) for its resilience to noise and ability to capture non-linear interactions between features [4].
- **Heuristic Optimization (CSP Logic):** A pure ML model might assign a low probability to a session with only one event. However, if that event is a System Logon (ID 4624), it constitutes definitive proof. We implemented hybrid logic to act as a "hard constraint" satisfaction mechanism:
- `if feats['critical_event_count'] > 0:`
- `    prob = max(prob, 0.95)`

  This ensures the system never misses a "Smoking Gun" artifact.

8. Model Evaluation & Comparison

The model was evaluated on a held-out test set of realistic, noisy data.

Metric	Value	Interpretation
Precision	**1.00**	**Zero False Positives.** The system never falsely accuses a user of being present when they were physically away.
Recall	**0.91**	**High Sensitivity.** The system detects 91% of active sessions, occasionally missing extremely stealthy usage but capturing the vast majority of activity.
F1 Score	**0.95**	An excellent balance between sensitivity and reliability, crucial for forensic applications.

9. Explainability & Visualization

To fulfill the "Ethical AI" requirement, we integrated **SHAP (SHapley Additive exPlanations)** [3]. A Waterfall Plot is generated for each prediction on the dashboard. This allows an investigator to see exactly *why* the AI flagged a session. For instance, it might display "+0.45 Probability due to High Action Diversity," proving that the decision was data-driven rather than a "black box" error.

10. Results & Discussion

The system successfully identified 100% of "Heavy Usage" cases and >90% of "Stealth Usage" cases. The "Heuristic Override" proved crucial in edge cases where there was a low volume of events but high significance, such as a single remote login. The combination of ML for pattern recognition and heuristics for definitive artifacts provided a robust defense against both false positives and false negatives.

11. Ethical AI & Limitations

- **Bias:** The model assumes "Active Presence" correlates with log volume. A user passively watching a downloaded movie might not generate many logs, potentially leading to a False Negative.
- **Transparency:** By using SHAP, we reduce the risk of "Algorithmic Bias" by revealing the decision factors to the human investigator, allowing for manual verification of the AI's logic [6].

12. Conclusion & Future Work

Veritas has demonstrated that AI can effectively augment digital forensics by combining pattern recognition with definitive heuristic logic. Future work includes the integration of **Live Memory Analysis (RAM dumps)** to detect artifacts not written to disk and the use of **Large Language Models (LLMs)** to generate natural language summaries of suspicious activity for non-technical stakeholders.

Github link: https://github.com/aayanrashid995/veritas-alibi-system

13. References

[1] S. M. Lundberg and S.-I. Lee, "A Unified Approach to Interpreting Model Predictions," in *Advances in Neural Information Processing Systems*, 2017, pp. 4765–4774.

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[5] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

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